

SP19-3: FC-2 Spectral Modularity Paper Outline

Working Title: “Spectral Modularity: BSD and Modularity as One Evaluation on D_{IV}^5 ” Lead: Lyra + Elie
Target: J. Funct. Anal. or Inventiones Mathematicae Date: May 13, 2026

Scope (honest, per Cal FC-5)

- This paper is about 49a1 specifically. Conductor $g^2=49$, CM by $Q(\sqrt{-7})$, torsion $Z/2Z$, rank 0.
- Do not generalize to “all modularity.” Wiles/BCDT remains external for existence of weight-2 newforms for arbitrary E/Q .
- The claim: For 49a1, modularity and BSD are one spectral evaluation — the Eisenstein residue at $s=1$ on $SO_0(5,2)$ at Wallach $k=2$ equals the BSD invariant $L(E,1)/\Omega = 1/\text{rank}$.

Sections (9)

1. Introduction (2-3pp)

State main theorem. Motivate: classical modularity and BSD are separate deep results; for 49a1 they are one spectral evaluation on D_{IV}^5 . Honest scope.

2. The Domain D_{IV}^5 and Its Parabolic Structure (3-4pp)

$D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$, rank 2, $n_C=5$. P_2 parabolic: Levi $M = GL(2) \times SO(3)$, $\dim N_P = g = 7$. Wallach set. K-type formula. Ring uniqueness (T1780). - T1762 (Toy 2091, 12/12), T1780 (Toy 2120, 10/10), T1829 (Toy 2151, 26/26). All D-tier.

3. The Eisenstein Series $E(f,s,P_2)$ and Its Pole at $s=1$ (4-5pp)

Langlands-Shahidi normalizing factor. Constant-term formula: $M(s,f) = \zeta(2s)L(2s-1, \text{Sym}^2 f) / [\zeta(2s+1) * L(2s, \text{Sym}^2 f)]$. For 49a1 (CM by $Q(\sqrt{-7})$): $\text{Sym}^2 f = \zeta L(\chi_{-7}) * L_K(\psi/\psi^\sigma)$. Pole from $\zeta(2s-1)$. Residue contains $L(1, \chi_{-7}) = \pi/\sqrt{7}$. - W-8b (Toy 2147, 10/10). D-tier.

4. The Residual Representation Is the Wallach π_2 (3-4pp)

Three-step identification: (1) K-type matching — lowest = trivial $SO(5) \times \text{weight-2}$, $\dim 1 = \pi_2$. (2) Holomorphicity + Harish-Chandra uniqueness — $J(f,1)$ is unique holomorphic irreducible with this K-type. (3) Non-vanishing via theta lift — $L(E,1) \neq 0$ for rank-0 49a1, so $\theta(f) \neq 0$ via Rallis. - FC-2a (Toy 2150, 15/15). D-tier.

5. The Rallis Inner Product Formula and the BSD Connection (3-4pp)

$\| \text{Res } E(f,s) \|^2 \sim L(E,1)L(1, \text{Ad } f)\text{vol}(G/K)$. BSD critical value $L(E,1)$ appears in residue norm. Combined with BSD formula: $L(E,1)/\Omega = 1/\text{rank} = \text{Wallach Plancherel ratio at } k=\text{rank}$. This is the unification. - T1756 (Paper #88), T1430 (1/rank universality). D-tier.

6. BST Integer Map of the Chain (2-3pp)

Every number in the Eisenstein $\rightarrow$ pole $\rightarrow$ residue $\rightarrow$ BSD chain is a BST integer. 20+ appearances, 0 unexplained. Systematic table: $\deg(r_1)=C_2$, $\dim N_P=g$, conductor= g^2 , CM disc= $-g$, $L(E,1)/\Omega=1/\text{rank}$, lowest K-type $\dim=1$, next= n_C . - Toys 2147, 2150. D-tier.

7. The F_1 Collapse and Structural Uniqueness (2-3pp)

At $q=1$: both sides collapse to rank=2 (T1808, D). Eichler-Shimura $T_1=2*\text{id}$. Poisson kernel at $q=1$ is identity. Chevalley extension uniqueness (T1809, C). Self-referential irreducibility $x^{g+x}\{N_c\}+1$ over F_2 (T1811, C). - T1808 (D), T1809 (C), T1810 (D), T1811 (C). Mixed D+C. - Gaps: T1809 functoriality step, T1811 formalization.

8. Honest Scope and Relation to Wiles/BCDT (1-2pp)

What IS claimed: For 49a1, modularity and BSD are one spectral evaluation. What is NOT: new proof of modularity for all E/Q ; Wiles/BCDT superseded; generalization beyond CM curves with conductor g^2 .

9. Conclusion and Future Directions (1p)

Extend to other CM curves (11a1, 37a1). FET conjecture (GC-17a). Wallach Universality connection.

Tier Assessment

Section	Tier	Key Evidence
2 (Domain)	D	T1762, T1780, T1829
3 (Eisenstein pole)	D	Toy 2147, 10/10
4 (Residual = π_2)	D	Toy 2150, 15/15
5 (Rallis + BSD)	D	T1756, T1430
6 (BST integers)	D	Toys 2147, 2150
7 (F_1 uniqueness)	D+C	T1808-T1811

Core chain (Sections 3-6): entirely D-tier. Uses only published theorems + verified computations.

Gaps for New Work

1. Toy 2133 (Elie, assigned): P_2 constant-term test for 49a1 + 11a1 + 37a1. Strengthens Section 3.
2. T1809 formalization: Chevalley extension for $(GL(2), SO(5,2))$ via Arthur. Finite computation.
3. T1811 formalization: Polynomial irreducibility $\rightarrow$ correspondence indecomposability. Needs rigorous proof.
4. Full leading coefficient: $L(E,1)/\Omega = 1/\text{rank}$ is proved. Full Bloch-Kato computation for completeness.
5. Non-archimedean verification table: a_p matching at small primes (partially in Toy 2147).

Key Files

- notes/BST_GC17b_Modularity_Via_Boundary_Invertibility.md — boundary-interior framework
- notes/BST_GC17c_Wallach_Point_Modularity_Seed.md — Wallach investigation
- notes/BST_GC17a_Modularity_Feasibility_Scoping.md — honest scope
- notes/BST_V1_Cuspidal_Test_GC17b.md — Cal V-1
- play/toy_2150_residual_rep_wallach_identification.py — FC-2a (15/15)
- play/toy_2147_eisenstein_wallach_factorization.py — W-8b (10/10)
- play/toy_2131_boundary_interior_modularity.py — GC-17b (35/35)